

Reasoning analysis — gemini-3.5-flash-lite (49 runs, real_gemini35flashlite_50)

Each model rationale is split into sentences; sentences making the same point across runs are grouped. **Shared** = the point appears in at least 60% of the runs in that group; **varying** = it appears in some runs but not most. Counts are runs, not sentences. 'Evidence cited' lists the numbers and times the model quoted, with the number of runs quoting each.

decision: Profile / basal 00:00

keep — 44/49 runs

Varying reasoning

- (5/44) Maintains stable overnight fasting glucose.
- (5/44) Dependencies: none.
- (3/44) Evidence shows minor dipping during local hours 00:00-01:00, but Oref1 low temp automation safely handles this without requiring basal rate reduction.
- (3/44) Overnight fasting glucose is generally stable; maintain current rate pending meal logging verification.
- (3/44) Overnight glucose is stable near target; evidence supports maintaining current basal delivery.
- (3/44) Uncertainty: unannounced evening activity.
- ... 7 more varying points

Points made by a single run only: 44.

Evidence cited: 00:00 (5), 0.6u/h (5), 01:00 (4), 02:00 (2), 18.5% (2), 106mg/dl (2), 93mg/dl (2), 70mg/dl (1), 100mg/dl (1).

change — 5/49 runs

Shared reasoning

- (3/5) Evidence shows 18.5% time below range at 01:00 local time.

Points made by a single run only: 6.

Phrases used by at least half the runs: “18.5 time below” (3), “time below range” (3), “basal rate” (3).

Evidence cited: 18.5% (3), 01:00 (2), 70mg/dl (1), 10.1% (1), 03:00 (1), 00:00 (1).

decision: Profile / basal 05:00

keep — 49/49 runs

Varying reasoning

- (9/49) Morning basal segment maintains stable early morning trends.
- (7/49) Dependencies: none.
- (5/49) Early morning glucose averages remain stable around 99 mg/dL.
- (5/49) Early morning stability is acceptable; no evidence of drift requiring adjustment.
- (4/49) Evidence: hourly glucose data.
- (3/49) Current early morning basal rate shows stable fasting control without excessive highs or lows.

- ... 6 more varying points

Points made by a single run only: 32.

Phrases used by at least half the runs: “early morning” (29).

Evidence cited: 99mg/dl (2), 0.7u/h (2), 07:00 (2), 97% (1), 89% (1), 98% (1), 110mg/dl (1), 05:00 (1), 100mg/dl (1), 106mg/dl (1), 03:00 (1).

decision: Profile / basal 06:00

keep — 49/49 runs

Varying reasoning

- (7/49) Dependencies: none.
- (7/49) Adequately covers morning rise and dawn phenomenon without excessive hypoglycemia.
- (4/49) Morning step-up matches natural dawn phenomenon patterns; keep current value pending structured observation.
- (4/49) Morning basal maintains stable fasting levels prior to morning rise.
- (3/49) Appropriate morning basal coverage supporting stable fasting glucose before morning rises.
- (3/49) Keep current pending meal-logging data.
- ... 13 more varying points

Points made by a single run only: 14.

Evidence cited: 0.8u/h (2).

decision: Profile / basal 10:00

keep — 49/49 runs

Varying reasoning

- (20/49) Mid-day basal segment maintains stable glucose prior to afternoon peaks.
- (6/49) Midday basal rate performs stably; afternoon elevations are driven by unannounced meals rather than basal deficit.
- (5/49) Uncertainty: unannounced meal timing.
- (5/49) Dependencies: none.
- (4/49) Maintains daytime stability.
- (3/49) Maintains stable midday glucose levels.
- ... 8 more varying points

Points made by a single run only: 14.

Evidence cited: 0.7u/h (2).

decision: Profile / basal 16:00

keep — 49/49 runs

Varying reasoning

- (11/49) Retain current setting; afternoon hyperglycemia is primarily driven by unannounced meal absorption rather than basal deficit.
- (5/49) Current rate of 0.5 U/h is appropriate; afternoon elevations are driven by unannounced meals rather than basal deficiency.
- (4/49) Uncertainty: exact meal timing.
- (3/49) Dependencies: none.
- (3/49) Afternoon basal rate is supported by active dynamic ISF and SMB automation handling postprandial absorption.
- (3/49) Dependencies: meal logging.
- ... 7 more varying points

Points made by a single run only: 32.

Evidence cited: 0.5u/h (2).

decision: Profile / basal 18:00

keep — 49/49 runs

Varying reasoning

- (13/49) Evening basal rate is appropriate for evening transition.
- (7/49) Dependencies: none.
- (6/49) Evening basal segment provides adequate transition into night.
- (6/49) Evening basal segment maintains acceptable stability.
- (5/49) Evidence: hourly glucose averages.
- (4/49) Evening basal segment supports stable pre-dinner glucose.
- ... 8 more varying points

Points made by a single run only: 11.

Phrases used by at least half the runs: “evening basal” (28).

Evidence cited: 0.6u/h (2), 00:00 (1).

decision: Profile / basal 19:00

keep — 49/49 runs

Varying reasoning

- (14/49) Late evening basal rate supports evening stability.
- (7/49) Dependencies: none.
- (5/49) Maintains stable evening baseline coverage prior to overnight fasting.
- (4/49) Supports evening requirements prior to overnight transition.
- (4/49) Maintains stability into the night.
- (4/49) Evidence: hourly glucose data.
- ... 10 more varying points

Points made by a single run only: 12.

Evidence cited: 0.7u/h (2).

decision: Profile / CR 00:00

keep — 49/49 runs

Varying reasoning

- (6/49) Carb ratios cannot be validated without recorded carb entries; keep current settings.
- (5/49) No direct carb entries to evaluate CR performance; keep current setting.
- (5/49) Evidence: zero carb entries.
- (4/49) Dependencies: carb logging.
- (4/49) Carb entries are absent; changing CR without meal data is unsupported.
- (3/49) Carb ratio cannot be evaluated without manual carb entry data.
- ... 12 more varying points

Points made by a single run only: 21.

Phrases used by at least half the runs: "carb entries" (29).

Evidence cited: 00:00 (1).

decision: Profile / CR 04:00

keep — 49/49 runs

Varying reasoning

- (5/49) No data contradicting current early-morning carb ratio.
- (5/49) Evidence: zero carb entries.
- (4/49) No structured carb data available to evaluate early morning CR performance.
- (4/49) Retain pending meal logging implementation.
- (3/49) Maintain current CR segment pending explicit meal logging.
- (3/49) Uncertainty: true CR adequacy.
- ... 13 more varying points

Points made by a single run only: 21.

Evidence cited: 04:00 (1).

decision: Profile / CR 08:00

keep — 49/49 runs

Varying reasoning

- (5/49) Evidence: zero carb entries.
- (5/49) No meal carb data available to evaluate morning CR performance.
- (3/49) Appropriate daytime carb ratio supporting robust closed-loop performance.
- (3/49) Keep current pending structured meal logging.
- (3/49) Maintain current CR segment pending explicit meal logging.
- (3/49) Morning CR segment supports breakfast window.
- ... 10 more varying points

Points made by a single run only: 29.

Evidence cited: 08:00 (1).

decision: Profile / CR 16:00

keep — 49/49 runs

Varying reasoning

- (5/49) Evidence: zero carb entries.
- (3/49) Afternoon spikes relate to unannounced intake rather than ratio defects.
- (3/49) Maintain current CR segment pending explicit meal logging.
- (3/49) Requires recorded meal carbohydrate data for audit.
- (3/49) Dependencies: meal logging.
- (3/49) Dependencies: carb logging.
- ... 11 more varying points

Points made by a single run only: 33.

Evidence cited: 16:00 (1).

decision: Profile / CR 20:00

keep — 49/49 runs

Varying reasoning

- (10/49) Evening CR segment maintains stable overnight transitions.
- (7/49) Evening carb ratio is performing within acceptable closed-loop parameters.
- (6/49) Keep current pending structured meal logging.
- (5/49) Evidence: zero carb entries.
- (3/49) Maintains evening carb ratio stability.
- (3/49) Dependencies: meal logging.
- ... 11 more varying points

Points made by a single run only: 15.

Evidence cited: 12.6g/u (1), 20:00 (1).

decision: Profile / ISF 00:00

keep — 48/49 runs

Varying reasoning

- (6/48) Flat ISF of 56 is actively modulated by Oref1 dynamic sensitivity (70%); changing static ISF without verified meal data is premature.
- (4/48) Flat profile ISF of 56 mg/dL/U is successfully supplemented by Dynamic ISF (70% adjustment factor) and Oref1 real-time scaling.
- (4/48) Changing base ISF without isolating meal effects is unjustified.
- (4/48) Uncertainty: dynamic scaling behavior.
- (4/48) Dependencies: automation settings.
- (3/48) Flat ISF of 56 mg/dL/U paired with Dynamic ISF (70%) provides safe, conservative insulin sensitivity scaling.
- ... 8 more varying points

Points made by a single run only: 35.

Phrases used by at least half the runs: “dynamic isf” (31), “flat isf” (25).

Evidence cited: 70% (27), 56mg/dl (12).

verify — 1/49 runs

Points made by a single run only: 1.

Phrases used by at least half the runs: “contributes to nocturnal” (1), “risk verify response” (1), “10 hour dia” (1), “flat isf combined” (1), “combined with 10” (1), “hour dia contributes” (1), “hypoglycemia risk verify” (1), “nocturnal hypoglycemia risk” (1), “response before altering” (1).

decision: Profile / target 00:00

keep — 48/49 runs

Varying reasoning

- (27/48) Overnight target of 90 mg/dL is appropriate for closed-loop operation.
- (5/48) Dependencies: none.
- (4/48) Keep current value.
- (3/48) Appropriate overnight target supporting fasting stability.
- (3/48) Evidence: stable overnight TIR.
- (2/48) Target of 90 mg/dL supports tight control without excessive hypoglycemia.
- ... 3 more varying points

Points made by a single run only: 19.

Phrases used by at least half the runs: “90 mg/dl” (25), “target of 90” (24), “overnight target” (24).

Evidence cited: 90mg/dl (25), 100mg/dl (1).

change — 1/49 runs

Points made by a single run only: 4.

Phrases used by at least half the runs: “basal sensitivity dependencies” (1), “18 time below” (1), “95 mg/dl reduces” (1), “individual basal sensitivity” (1), “90 mg/dl target” (1), “reduces nocturnal hypoglycemia” (1), “early night hours” (1), “nocturnal hypoglycemia risk” (1), “mg/dl target raising” (1), “11 18 time” (1), “hypoglycemia risk uncertainty” (1), “sensitivity dependencies safety” (1).

Evidence cited: 18% (1), 95mg/dl (1), 11% (1), 90mg/dl (1).

decision: Profile / target 08:00

keep — 49/49 runs

Varying reasoning

- (22/49) Daytime target of 100 mg/dL provides a safe buffer against postprandial variability.
- (6/49) Dependencies: none.
- (6/49) Daytime target supports safe closed-loop operation.
- (5/49) Appropriate daytime target.
- (3/49) Appropriate daytime target balancing safety and tight control.
- (3/49) Daytime target is appropriate.
- ... 4 more varying points

Points made by a single run only: 15.

Phrases used by at least half the runs: “daytime target” (43), “100 mg/dl” (25).

Evidence cited: 100mg/dl (25).

decision: Profile / target 23:00

keep — 49/49 runs

Varying reasoning

- (13/49) Late evening target is appropriate.
- (12/49) Late evening target of 90 mg/dL aligns with overnight target.
- (7/49) Dependencies: none.
- (4/49) Nighttime target aligns with closed-loop goals.
- (3/49) Night target transition is well-configured.
- (3/49) Maintains safe evening target.
- ... 4 more varying points

Points made by a single run only: 19.

Phrases used by at least half the runs: “evening target” (25).

Evidence cited: 90mg/dl (10), 00:00 (2).

decision: Profile / DIA

keep — 49/49 runs

Varying reasoning

- (17/49) 10-hour DIA provides conservative insulin stacking protection in closed loop.
- (8/49) 10-hour DIA provides conservative insulin-on-board tracking suitable for closed-loop safety.
- (5/49) Dependencies: none.
- (3/49) Dependencies: safety limits.
- (3/49) DIA of 10 hours matches insulin pharmacokinetics and supports safe IOB calculation under oref1.
- (3/49) Uncertainty: insulin kinetics.
- ... 5 more varying points

Points made by a single run only: 23.

Phrases used by at least half the runs: “10 hour” (37), “10 hour dia” (34).

Evidence cited: 10h (5).

decision: AAPS / SMB and UAM

keep — 49/49 runs

Varying reasoning

- (26/49) Both SMB and UAM are essential for managing unannounced meals given the lack of manual carb entries.
- (7/49) Dependencies: none.
- (4/49) Essential for managing unannounced meals given zero logged carb entries.
- (4/49) Automation features are functioning correctly to manage unannounced excursions.
- (3/49) Both SMB and UAM are appropriately enabled to support unannounced meal handling.
- (2/49) Essential for automated closed-loop management of unannounced intake in the absence of manual carb entries.
- ... 5 more varying points

Points made by a single run only: 12.

Evidence cited: 86% (1).

decision: AAPS / SMB activation conditions

keep — 49/49 runs

Varying reasoning

- (8/49) Activation conditions appropriately support unannounced meal detection and micro-bolusing.
- (7/49) Dependencies: none.
- (5/49) Appropriately configured for unannounced meal management.
- (4/49) Current activation settings support UAM-driven micro-bolusing in the absence of carb entries.
- (4/49) Appropriate configuration for an unannounced-meal operating style.
- (3/49) Activation settings correctly support closed-loop responsiveness to unannounced rises.

- ... 8 more varying points

Points made by a single run only: 19.

decision: AAPS / SMB strength and frequency

keep — 49/49 runs

Varying reasoning

- (7/49) 3-minute interval with 15 min max SMB basal and 20 min max UAM basal provides responsive microbolusing.
- (7/49) Provides responsive micro-bolusing without excessive aggressiveness.
- (6/49) Uncertainty: none.
- (6/49) Current delivery intervals and caps provide responsive loop adjustments without excessive insulin stacking.
- (6/49) Delivery intervals and limits are correctly configured for oref1 closed-loop operation.
- (3/49) Delivery limits are safe and responsive.
- ... 7 more varying points

Points made by a single run only: 18.

Evidence cited: 20min (2), 15min (1).

decision: AAPS / max basal and max IOB limits

keep — 41/49 runs

Varying reasoning

- (7/41) Safety ceilings are appropriately configured.
- (7/41) Safety limits provide adequate boundaries for automated delivery.
- (6/41) Dependencies: none.
- (3/41) Max IOB of 25 U is high relative to TDD (~20 U/day) but requires careful verification before alteration.
- (3/41) Safety limits are functional, though max IOB should be reviewed against TDD in future optimization.
- (3/41) Safe upper safety ceilings that do not restrict normal loop operation.
- ... 6 more varying points

Points made by a single run only: 21.

Evidence cited: 25u (6), 20u (3), 12u/h (2), 5.9u (1), 6u (1).

change — 8/49 runs

Varying reasoning

- (4/8) Max IOB of 25 U is excessively high for total daily insulin (~20 U/day) and poses a severe stacking risk contributing to hypoglycemia.
- (2/8) Max IOB of 25 U is excessively high for a ~20 U daily dose, permitting dangerous insulin stacking that contributes to nocturnal hypoglycemia.
- (2/8) Reducing Max IOB to 6 U provides essential protection against insulin stacking.

Points made by a single run only: 5.

Phrases used by at least half the runs: “max iob” (8), “iob of 25” (7), “25 is excessively” (5), “daily insulin” (5), “excessively high” (5), “insulin stacking” (5), “reducing max iob” (5), “total daily” (4).

Evidence cited: 25u (8), 20u (5), 6u (3), 10u (1), 8u (1), 15u (1).

decision: AAPS / Autosens and Dynamic ISF

keep — 49/49 runs

Varying reasoning

- (10/49) Dynamic ISF successfully compensates for the flat profile ISF.
- (6/49) Dependencies: none.
- (6/49) Dynamic ISF adjustment factor of 70% appropriately scales sensitivity for real-time glycemic changes.
- (5/49) Dynamic ISF at 70% provides necessary adaptation to real-time glucose kinetics.
- (4/49) Dynamic ISF at 70% effectively scales sensitivity without requiring autosens target changes.
- (3/49) Dynamic ISF with 70% adjustment factor is appropriately active.
- ... 8 more varying points

Points made by a single run only: 17.

Phrases used by at least half the runs: “dynamic isf” (37).

Evidence cited: 70% (27).

decision: AAPS / carbohydrate absorption model

keep — 49/49 runs

Varying reasoning

- (7/49) Carb absorption parameters align with standard Oref1 defaults.
- (6/49) Uncertainty: none.
- (6/49) Carb impact and absorption limits appropriately support UAM detection.
- (6/49) 7-hour absorption window appropriately accommodates delayed gastric emptying from GLP-1 therapy.
- (5/49) 7-hour maximum absorption window properly matches GLP-1 delayed gastric emptying kinetics.
- (3/49) 7-hour max absorption correctly accommodates delayed gastric emptying from GLP-1 therapy.
- ... 7 more varying points

Points made by a single run only: 19.

Evidence cited: 5min (2), 8mg/dl (2).

profile recommendation

basal — 41/49 runs

Varying reasoning

- (20/41) Keep current profile settings unchanged while initiating trial manual carbohydrate logging.
- (9/41) Overall time in range is strong at 86.0%, and closed-loop automation (Oref1 with dynamic sensitivity) is actively managing variability.
- (6/41) Safely reduce the 00:00 basal segment from 0.6 U/h to 0.5 U/h.
- (3/41) Overnight basal rates maintain stable fasting glucose near target, and existing profile settings achieve 86.0% TIR.
- (3/41) Current profile basal segments, ISF, CR, and target values maintain a strong 86% TIR.
- (3/41) Keep current profile settings while closely monitoring nocturnal glucose stability and afternoon postprandial excursions.
- ... 7 more varying points

Points made by a single run only: 50.

Phrases used by at least half the runs: “keep current profile” (31).

Evidence cited: 86% (14), 86.0% (12), 5.1% (10), 00:00 (6), 0.6u/h (5), 0.5u/h (5), 70mg/dl (4), 02:00 (2), 18.5% (2), 98% (1), 01:00 (1), 06:00 (1).

isf — 3/49 runs

Points made by a single run only: 9.

Phrases used by at least half the runs: “keep current profile” (3), “profile parameters” (2), “risks introducing” (2).

Evidence cited: 86% (1), 56mg/dl (1).

dia — 2/49 runs

Points made by a single run only: 5.

Phrases used by at least half the runs: “meal carbohydrate” (2), “keep current profile” (2), “carbohydrate logging” (2), “profile settings including” (2), “logging data risks” (1), “verification changing insulin” (1), “parameters without carbohydrate” (1), “automation actively manages” (1), “meal carbohydrate entries” (1), “dia and flat” (1), “introducing further variability” (1), “inputs to optimize” (1).

Evidence cited: 10h (1), 56mg/dl (1), 86% (1).

cr — 2/49 runs

Points made by a single run only: 6.

Phrases used by at least half the runs: “establishing baseline” (2), “maintain an 86” (2), “keep current profile” (2), “automation changing foundational” (1), “86 time” (1), “first establishing baseline” (1), “isf parameters prematurely” (1), “prematurely without first” (1), “manual carb logging” (1), “oref closed loop” (1), “conducting structured logging” (1), “changing foundational basal” (1).

Evidence cited: 86% (2).

target — 1/49 runs

Points made by a single run only: 2.

Phrases used by at least half the runs: “90 mg/dl target” (1), “target segment low” (1), “range reaches 18.5” (1), “00 target segment” (1), “18.5 at 01” (1), “mg/dl will improve” (1), “low and high” (1), “segment from 90” (1), “mg/dl target raising” (1), “mg/dl overnight time” (1), “00 local time” (1), “high from 90” (1).

Evidence cited: 00:00 (1), 95mg/dl (1), 18.5% (1), 90mg/dl (1), 01:00 (1).

primary recommendation

basal — 22/49 runs

Varying reasoning

- (11/22) Overnight glucose levels show consistent dipping and time below range between 00:00 and 02:00 local time (reaching 18.5% below range at 01:00).
- (9/22) Verify Overnight Basal Rates.
- (4/22) Overnight and early morning glucose values show frequent dipping and time-below-range up to 18.5%.
- (4/22) Monitor Overnight Basal and Hypoglycemia.
- (3/22) Reduce Overnight Basal Rates.
- (2/22) Withhold basal changes until carbohydrate logging is introduced to determine if lows are driven by basal over-delivery or unannounced meal absorption tails.
- ... 1 more varying points

Points made by a single run only: 20.

Phrases used by at least half the runs: “below range” (17), “time below range” (16), “overnight basal” (15), “basal rates” (11).

Evidence cited: 18.5% (14), 00:00 (9), 02:00 (7), 01:00 (6), 0.6u/h (4), 03:00 (3), 0.8u/h (3), 10% (2), 5.1% (2), 70mg/dl (2), 56mg/dl (1), 90mg/dl (1).

meal — 7/49 runs

Varying reasoning

- (4/7) Introduce Manual Carbohydrate Logging.

Points made by a single run only: 13.

Phrases used by at least half the runs: “carbohydrate logging” (6).

Evidence cited: 293mg/dl (1).

cr — 6/49 runs

Varying reasoning

- (3/6) Telemetry shows zero explicit carb entries, forcing closed-loop automation to rely entirely on UAM and reactive microbolusing.
- (3/6) Initiate Carbohydrate Logging.
- (2/6) Recording carb entries for meals will allow proper prebolusing and reduce reliance on reactive stacking.

Points made by a single run only: 7.

Phrases used by at least half the runs: “carbohydrate logging” (3), “uam and reactive” (3), “entirely on uam” (3), “closed loop” (3).

smb — 5/49 runs

Shared reasoning

- (4/5) Review Max IOB Safety Limit.

Varying reasoning

- (2/5) Max IOB is set to 25 U against a total daily dose of approximately 20 U/day.

Points made by a single run only: 5.

Phrases used by at least half the runs: “max iob” (4), “iob safety limit” (3), “max iob limit” (3), “total daily dose” (3).

Evidence cited: 25u (4), 20u (3).

aaps — 5/49 runs

Shared reasoning

- (5/5) Max IOB of 25 U is excessively high for a total daily insulin intake of approximately 20 U/day, offering inadequate restriction against insulin stacking.
- (5/5) Reduce Max IOB Safety Limit.

Points made by a single run only: 2.

Phrases used by at least half the runs: “daily insulin” (5), “max iob” (5), “insulin stacking” (5), “iob safety limit” (4), “total daily insulin” (4), “20 u/day” (3), “reduce max iob” (3), “max iob safety” (3).

Evidence cited: 25u (5), 20u (3), 6u (1), 8u (1).

meal_strategy — 1/49 runs

Points made by a single run only: 2.

Phrases used by at least half the runs: “amounts for meals” (1), “logging log carbohydrate” (1), “consistent meal logging” (1), “counting and pre” (1), “late insulin stacking” (1), “establish consistent meal” (1), “log carbohydrate amounts” (1), “reactive uam” (1), “consistently to enable” (1), “enable proper aaps” (1), “reliance on reactive” (1), “proper aaps carb” (1).

profile — 1/49 runs

Points made by a single run only: 2.

Phrases used by at least half the runs: “meal carb entries” (1), “profile settings unchanged” (1), “evaluate true operational” (1), “establishing baseline” (1), “reduce afternoon postprandial” (1), “afternoon postprandial peaks” (1), “exacerbating overnight hypoglycemia” (1), “manual meal carb” (1), “cr and reduce” (1), “unchanged while establishing” (1), “verify meal logging” (1), “entries to evaluate” (1).

target — 1/49 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “adjust overnight target” (1), “90 mg/dl target” (1), “00 to 08” (1), “early night hours” (1), “slightly can reduce” (1), “08 00 contributes” (1), “11 18 hypoglycemia” (1), “target from 00” (1), “raising the overnight” (1), “target the 90” (1), “night hours raising” (1), “reduce nighttime lows” (1).

Evidence cited: 11% (1), 08:00 (1), 00:00 (1), 18% (1), 90mg/dl (1).

safety — 1/49 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “anomalies while remaining” (1), “above normal operational” (1), “user total daily” (1), “set to 25” (1), “review max iob” (1), “safety limit max” (1), “iob safety limit” (1), “limit max iob” (1), “remaining well above” (1), “max iob safety” (1), “dose 20 u/day” (1), “total daily dose” (1).

Evidence cited: 25u (1), 20u (1).

summary

49 runs

Shared reasoning

- (34/49) A 14-day analysis of Nightscout telemetry and AAPS profile data shows an overall Time in Range of 86.0% with Time Above Range at 8.9% and Time Below Range at 5.1%.

Varying reasoning

- (15/49) Analysis of 14 days of Nightscout telemetry and AAPS closed-loop data shows strong overall glycemic control with an 86.0% Time in Range.
- (11/49) The user operates an AAPS closed-loop system utilizing SMB and UAM with a flat ISF (56 mg/dL/U), a 10-hour DIA, and dynamic sensitivity enabled.
- (6/49) The user operates without manual carbohydrate entries, relying entirely on unannounced meal detection.
- (5/49) The closed-loop system relies heavily on UAM and SMB automation with zero explicit carbohydrate entries recorded in the treatment summary.
- (3/49) However, Time Below Range reaches 5.1%, driven by occasional overnight and post-correction dips into the 40-50 mg/dL range.
- (3/49) The dataset contains no recorded carbohydrate treatment entries, indicating that meals are unannounced or unlogged, relying entirely on UAM and reactive micro-bolusing to manage postprandial glycemic excursions.
- ... 7 more varying points

Points made by a single run only: 55.

Phrases used by at least half the runs: “time in range” (49), “below range” (34), “closed loop” (31), “time below range” (31), “nightscout telemetry” (30), “14 days” (28), “86.0 time” (28), “telemetry and aaps” (27), “days of nightscout” (25).

Evidence cited: 86.0% (48), 5.1% (42), 56mg/dl (20), 8.9% (17), 70mg/dl (13), 32.6% (8), 180mg/dl (7), 121.6mg/dl (6), 98.7% (4), 25u (4), 70% (3), 18.5% (3).

issues

49 runs

Varying reasoning

- (15/49) Recurrent hypoglycemia excursions (5.1% time below range) particularly during overnight and early morning hours (local hours 00:00-02:00 up to 18.5% below range).
- (15/49) Time below range (5.1%) exceeds the standard 4% safety threshold, driven by nocturnal and early-morning dipping.
- (10/49) Complete absence of logged carbohydrate entries (0 entries), forcing the closed-loop system to rely entirely on reactive UAM.
- (7/49) Afternoon postprandial hyperglycemia peaks (time above range reaching 32.1% during local hour 16:00).
- (7/49) Excessively high safety limits (Max IOB 25 U and Max Basal 12 U/h) relative to total daily insulin (~20 U/day).
- (5/49) Postprandial hyperglycemia peaks in the late afternoon (local hours 15:00–17:00).
- ... 12 more varying points

Points made by a single run only: 31.

Phrases used by at least half the runs: “time below” (39), “below range” (37), “time below range” (35).

Evidence cited: 5.1% (42), 70mg/dl (22), 56mg/dl (10), 18.5% (9), 4% (9), 16:00 (8), 25u (8), 00:00 (7), 32.1% (7), 15:00 (7), 02:00 (6), 17:00 (6).

step findings: data_quality

49 runs

Varying reasoning

- (22/49) CGM coverage is 98.7% with 3,995 valid entries over 14 days.
- (9/49) Telemetry is complete with extensive loop statuses and treatment records, though manual carb entries are entirely absent.
- (7/49) CGM coverage is excellent at 98.7% over 3,995 valid entries.
- (5/49) Data quality status is good with 10 minor gaps (largest 65 minutes).
- (3/49) Data quality is high and fully sufficient for closed-loop telemetry and profile analysis.
- (2/49) Data quality is good and fully supports chronological and aggregate loop telemetry analysis, though explicit meal carbohydrate entries are completely absent.
- ... 5 more varying points

Points made by a single run only: 25.

Phrases used by at least half the runs: “cgm coverage” (27).

Evidence cited: 98.7% (37), 65min (5), 65.0min (1), 5min (1).

step findings: safety_overview

49 runs

Varying reasoning

- (22/49) Time in Range is 86.0%, Time Above Range is 8.9%, and Time Below Range is 5.1%.
- (3/49) Time in Range is 86.0% and average glucose is 121.6 mg/dL.
- (3/49) Hypoglycemia occurs primarily overnight, while hyperglycemia peaks in the afternoon.
- (2/49) Strong overall time in range (86.0%), but significant time below range (5.1%) with glucose dropping to 41-42 mg/dL on multiple days, imposing a strict safety constraint against aggressive insulin increases.
- (2/49) Overall glycemic control is favorable at 86.0% TIR, but 5.1% time below range indicates significant hypoglycemia risk, particularly in late-night and early-morning hours.
- (2/49) Time in range is 86.0%, but time below range reaches 5.1% with lows down to 41–42 mg/dL, primarily between local 01:00 and 06:00.
- ... 2 more varying points

Points made by a single run only: 32.

Phrases used by at least half the runs: “below range” (34), “time below range” (31).

Evidence cited: 5.1% (44), 86.0% (30), 8.9% (18), 4% (12), 70mg/dl (8), 293mg/dl (6), 42mg/dl (6), 121.6mg/dl (4), 50mg/dl (3), 86% (3), 32.6% (2), 56mg/dl (2).

step findings: basal_targets

49 runs

Varying reasoning

- (16/49) Overnight basal rates maintain stable fasting levels around 90-100 mg/dL but contribute to early morning dips (up to 18.5% time below range at 01:00 local time).
- (5/49) Overnight basal rates show dipping below range during local hours 00:00-02:00, while afternoon hours exhibit postprandial elevation.
- (4/49) Overnight glucose dips during local hours 00:00–02:00 while late afternoon hours 15:00–17:00 show average glucose rising above 148–162 mg/dL.

- (3/49) Afternoon elevations reflect unannounced meals.
- (3/49) Overnight basal rates combined with flat ISF contribute to nocturnal hypoglycemia risk, while afternoon elevations reflect postprandial or resistance periods.
- (2/49) Overnight basal delivery (00:00-04:00) and late-afternoon basal delivery interact with active insulin to cause micro-dips below 70 mg/dL.
- ... 5 more varying points

Points made by a single run only: 21.

Phrases used by at least half the runs: “overnight basal” (34), “basal rates” (25).

Evidence cited: 00:00 (16), 18.5% (13), 02:00 (11), 01:00 (10), 70mg/dl (6), 162.8mg/dl (4), 16:00 (4), 162mg/dl (4), 18% (4), 0.8u/h (4), 0.7u/h (4), 0.6u/h (3).

step findings: profile_cr_isf_dia

49 runs

Varying reasoning

- (27/49) Profile ISF is flat at 56 mg/dL/U and DIA is 10 hours.
- (2/49) Static profile ISF of 56 mg/dL/U is paired with dynamic ISF (70% adjustment factor) and a 10-hour DIA.
- (2/49) Static profile ISF is set to 56 mg/dL/U and modulated by dynamic sensitivity, but lack of carb entries blocks direct CR validation.
- (2/49) The flat ISF combined with aggressive SMBs and dynamic ISF contributes to insulin stacking during sensitive overnight periods.
- (2/49) Dynamic ISF (70% adjustment factor) actively scales sensitivity during high insulin demand, interacting with SMB automation.
- (2/49) Flat profile ISF of 56 mg/dL/U and 10h DIA rely heavily on dynamic sensitivity and closed-loop automation.
- ... 3 more varying points

Points made by a single run only: 34.

Phrases used by at least half the runs: “56 mg/dl/u” (37).

Evidence cited: 56mg/dl (37), 70% (16), 10h (14), 13.2g/u (11), 100mg/dl (1), 24h (1).

step findings: automation_smb_uam

49 runs

Varying reasoning

- (13/49) SMB and UAM are enabled with smb_always_enabled=true.
- (5/49) The loop relies entirely on UAM and SMB automation to handle postprandial excursions due to missing carb entries.
- (4/49) SMB and UAM are fully enabled with smb_always_enabled=true, supporting an unannounced meal strategy.
- (4/49) SMB and UAM enabled with SMB always enabled, compensating for unannounced meals in the absence of declared carbs.
- (2/49) Automation heavily compensates for unannounced meals, but lack of manual carb entry increases reliance on reactive UAM dosing.
- (2/49) SMB and UAM are enabled with smb_always_enabled set to true and smb_with_cob_enabled set to false, matching an unannounced meal operating model.
- ... 1 more varying points

Points made by a single run only: 31.

Phrases used by at least half the runs: “smb and uam” (36).

step findings: automation_sensitivity_limits

49 runs

Varying reasoning

- (21/49) Autosens is disabled while Dynamic ISF is active (70%).
- (18/49) Safety limits (max basal 12 U/h, max IOB 25 U) provide wide safety ceilings.
- (5/49) Max IOB of 25 U is exceptionally high relative to total daily insulin (~20 U/day), creating severe insulin stacking risk that correlates with observed hypoglycemia.
- (3/49) Safety limits (max basal 12 U/h, max IOB 25 U) provide wide upper ceilings without restricting normal loop operation.
- (2/49) Autosens is disabled while Dynamic ISF is active at 70%, tightening effective ISF and increasing microbolus potency during downward trends.
- (2/49) Static Autosens is disabled while dynamic sensitivity is active at 70%, allowing real-time ISF adaptation.

Points made by a single run only: 23.

Phrases used by at least half the runs: “max iob” (32), “dynamic isf” (31).

Evidence cited: 25u (34), 70% (33), 12u/h (23), 20u (7), 6u (1).

step findings: meal_bolus_strategy

49 runs

Varying reasoning

- (4/49) No explicit meal boluses or carbohydrate entries are recorded.
- (3/49) Without manual carb entries, meals rely entirely on UAM and SMB reactive microbolusing, leading to predictable afternoon postprandial peaks.
- (2/49) GLP-1 therapy is administered periodically (1.25mg), contributing to gastric emptying delays.
- (2/49) In the absence of manual carb entries, the system relies exclusively on UAM and closed-loop adjustments, which maintains good TIR but permits afternoon postprandial excursions.
- (2/49) Without manual carb entries or explicit meal bolus declarations, insulin delivery relies entirely on Oref1 UAM detection, microboluses, and manual correction boluses.
- (2/49) Lack of meal announcements leads to reactive dosing spikes followed by late-window hypoglycemia risk when insulin activity overlaps with high sensitivity.
- ... 6 more varying points

Points made by a single run only: 33.

Evidence cited: 17:00 (1), 14:00 (1), 162mg/dl (1).

step findings: meal_ecarbs_absorption

49 runs

Varying reasoning

- (12/49) Extended carbs and e-carbs are not utilized.
- (8/49) No e-carbs or extended carb entries are utilized in the dataset.
- (6/49) GLP-1 therapy notes (1.25mg on Aug 14, 17, and 20) indicate delayed gastric emptying, which contributes to prolonged postprandial glucose tails.
- (5/49) The system relies entirely on UAM absorption modeling.

- (4/49) Oref1 relies on standard UAM settings with a 7-hour meal max absorption window, which appropriately accommodates GLP-1 delayed absorption.
- (4/49) Absorption modeling relies on default Oref1 dynamic carb impact detection and UAM.
- ... 9 more varying points

Points made by a single run only: 11.

Evidence cited: 5min (7), 8mg/dl (7), 7h (3).

step findings: synthesis_plan

49 runs

Varying reasoning

- (3/49) The primary recommendation is to keep current profile settings while establishing a consistent meal logging baseline to improve closed-loop performance.
- (2/49) Secondary issue is afternoon hyperglycemia from GLP-1 delayed absorption.
- (2/49) Data supports keeping profile and automation settings stable while initiating formal carbohydrate logging and meal prebolusing to reduce reliance on reactive UAM and mitigate late-night hypoglycemia.
- (2/49) Sequential verification required: first reduce max IOB safety limits to prevent insulin stacking, then evaluate manual meal carb logging.
- (2/49) Overnight hypoglycemia and afternoon hyperglycemia can be addressed by sequential profile verification and optional meal logging before adjusting insulin parameters.
- (2/49) Prioritizing a reduction in max IOB and verifying basal/ISF stability before making further automated adjustments will safely mitigate hypoglycemia risk.
- ... 2 more varying points

Points made by a single run only: 61.

Evidence cited: 5.1% (8), 86% (3), 10h (2), 70mg/dl (2), 25u (2), 90mg/dl (1), 56mg/dl (1), 00:00 (1).

meal strategy

49 runs

Varying reasoning

- (9/49) In the complete absence of declared meal carbohydrate entries (0g logged), closed-loop automation must rely entirely on UAM and SMB.
- (4/49) Maintain UAM-driven closed-loop automation (smb_enabled=true, uam_enabled=true, smb_always_enabled=true) with the 7-hour absorption model.
- (3/49) Maintain the current automated UAM and SMB delivery model while piloting manual carbohydrate logging for high-carbohydrate meals to reduce afternoon postprandial hyperglycemia peaks.
- (2/49) Maintain closed-loop UAM and SMB automation with a 3-minute interval, active Dynamic ISF (70%), and a 7-hour max absorption window, supplemented by optional manual meal announcements for known carbohydrate loads to smooth afternoon postprandial plateaus.
- (2/49) Introduce consistent manual carbohydrate entry and modest prebolusing for meals to give the closed-loop algorithm advance notice, reducing reliance on reactive UAM microboluses and smoothing out post-prandial spikes.
- (2/49) Maintain current closed-loop automation settings (SMB enabled, UAM enabled, Dynamic ISF active) while introducing consistent manual carbohydrate logging ahead of meals to enable COB prediction and pre-bolusing.
- ... 13 more varying points

Points made by a single run only: 87.

Phrases used by at least half the runs: "closed loop" (26).

Evidence cited: 70% (5), 20min (3), 86% (3), 15min (2), 86.0% (2), 4h (1), 32% (1), 5.1% (1), 8u (1), 100% (1), 7h (1), 162mg/dl (1).